Table 1. Candidate domains of tumor-associated brain frailty, instruments identified in the core evidence base, and remaining gaps.

Domain	Studies with data (n)	Instruments or measures identified	Direction of the evidence	Main limitation
1. Systemic frailty	7	mFI-5 (3)[34-36], mFI-11 (1)[37], Hopkins Frailty Score (1)[38], Risk Analysis Index (1)[36], Fried phenotype (1)[39], comorbidity count (1)[40]	Higher frailty linked to worse outcome or improved by rehabilitation in 7 of 7	General-surgery indices, no brain-specific item, thresholds differ, no glioma-only or meningioma-only sample
2. Cognitive	30 (plus 1 protocol)	Multidomain batteries (14), domain-specific tests (4), brief screeners (7), computerized batteries (4), patient-reported scale (1). MMSE in 5, MoCA in 2 with data, no MoCA-INA	Impaired before surgery in 8 of 8 studies that tested it. After surgery, among 18 cohorts, improved 6, no change 3, decline 3, transient decline 3, others 3	No shared tool or time point, early decline in glioma cohorts, improvement in meningioma cohorts, MMSE-J missed deficits in one study
3. Neurological	16	Binary new deficit (8), Barthel index (5), GOS or GOSE (3), KPS (2), mRS (3), GCS (2), NIHSS (1), House-Brackmann (1), intraoperative MEP (2). No NANO	Deficit rates 6.7% to 50% depending on definition	Nine instruments or definitions, none designed for neuro-oncology
4. Imaging	32	DWI for ischemic lesions (6), volumetric MRI (13), perfusion or permeability (4), functional connectivity (7), CT findings (7), contrast-enhanced ultrasound (1)	New DWI lesions after 18.8% to 45.7% of resections	Imaging describes tumor or injury, no brain frailty score, no CT-based or muscle-based metric

Counts are studies, and a study can contribute to more than one row. CT, computed tomography. DWI, diffusion-weighted imaging. GCS, Glasgow Coma Scale. GOS, Glasgow Outcome Scale. GOSE, extended Glasgow Outcome Scale. KPS, Karnofsky Performance Status. mFI-5, five-factor modified frailty index. mFI-11, eleven-item modified frailty index. MEP, motor evoked potential. MMSE, Mini-Mental State Examination. MoCA, Montreal Cognitive Assessment. MoCA-INA, Indonesian version of the MoCA. mRS, modified Rankin Scale. NANO, Neurologic Assessment in Neuro-Oncology. NIHSS, National Institutes of Health Stroke Scale.

Table 2. Answers to the seven review questions and the status of the mapped evidence.

Question	Answer from the mapped evidence	Basis	Status
RQ1. How is frailty or brain frailty defined and measured?	No brain-specific definition. Frailty is borrowed through seven general indices, in mixed tumor samples, with differing thresholds	7 core studies	Answered, and the answer is a gap
RQ2. Which domains and instruments characterize it?	Four domains map onto existing studies (systemic, cognitive, neurological, imaging). Each has several instruments and no shared standard	7, 30, 16, and 32 core studies	Answered descriptively
RQ3. Is neuroinflammation present in benign as well as malignant tumors?	Documented in glioma tissue, cerebrospinal fluid, and microdialysate. Benign tumor evidence rests on external reviews, and meningioma-only studies show injury and vulnerability but no cytokines	3 glioma or mixed studies, 0 meningioma	Partly answered
RQ4. Which biomarkers can detect neuroinflammation and injury?	S100B, NSE, GFAP, tau, and NfL track injury. Cytokines require tissue, cerebrospinal fluid, or microdialysis. No validated neuroinflammation panel	12 injury and 7 inflammatory studies	Partly answered
RQ5. What supports perioperative brain injury as a second hit?	New DWI lesions after 18.8% to 45.7% of resections, marker rise in 4 of 4 studies, delirium in 4.2% to 35%. Interaction with preoperative state untested	Supplemental Digital Content 10	Partly answered
RQ6. Which instruments exist, including MoCA-INA and NANO-Ina?	None of the core studies used either. Six instrument-focused studies document both, without perioperative testing	0 core, 6 instrument-focused	Answered, with a gap
RQ7. Where are the gaps?	No definition, no brain-specific index, no CT or muscle metric, no meningioma cytokine data, no interaction test, no perioperative validation of the Indonesian instruments	All of the above	Answered

Status describes how fully the mapped evidence answers the question. Partly answered and gap findings are reported as results of a scoping review. DWI, diffusion-weighted imaging. GFAP, glial fibrillary acidic protein. MoCA-INA, Indonesian version of the Montreal Cognitive Assessment. NANO-Ina, Indonesian version of the Neurologic Assessment in Neuro-Oncology scale. NfL, neurofilament light chain. NSE, neuron-specific enolase. RQ, review question.